NEA/HS/6.6/FOR01

20 Dec 2024
Updated 21 May 2025 – test method for water-based paints
Updated 9 Feb 2026 – 'Information to be included in test reports' added under Annex B

To: Licence Holders, Industry Associations, Traders & Declaring Agents

Dear Sir/Madam

CONTROL OF FORMALDEHYDE IN PAINTS UNDER THE ENVIRONMENTAL PROTECTION AND MANAGEMENT ACT

With effect from 1 Jan 2026, the National Environment Agency ("**NEA**") will be implementing the following control measures for formaldehyde in paints under the Environmental Protection and Management Act ("**EPMA**"):

- The import, manufacture, and/or sale for local consumption of interior paints[1] containing ≥ 0.01% weight in weight of formaldehyde will be prohibited in Singapore.

- Paints imported, manufactured, and/or sold for local consumption in Singapore for outdoor application or industrial use [2] containing ≥ 0.01% weight in weight of formaldehyde will need to comply with the labelling requirements specified in Annex A.

2 The above requirements will not apply to paints which are imported for re-export or manufactured for export purposes. However, a Hazardous Substance Licence will be required for companies that import and/or manufacture interior paints containing ≥ 0.01% weight in weight of formaldehyde for export or re-export purposes.

Submission of test reports

3 Accordingly, importers and manufacturers of interior paints will have to submit test reports to NEA upon request that indicate the measured total in-can formaldehyde content in each interior paint product is less than 0.01% weight in weight. Such test reports will have to be prepared by laboratories accredited by accreditation bodies which are signatories to the International Laboratory Accreditation Cooperation Mutual Recognition Arrangement (ILAC MRA). Please see Annex B for more information on the accepted test methods and information to be included in test reports.

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[1] Paints applied on interior surfaces such as internal walls and ceilings of buildings.
[2] Paints for industrial use refer to paints for industrial application, including but not limited to those conferring anti-fouling properties for marine vessels and anti-corrosion properties for metal structures and equipment.

Harmonized System ("HS") Codes and Product Codes

4 The list of new HS Codes and Product Codes for paints containing formaldehyde are listed in Annex C. With effect from 1 Jan 2026, traders are required to submit import and export TradeNet permit applications using the appropriate HS Codes and Product Codes. All such TradeNet permit applications will be subject to Chemical Control and Management Department (CCMD)'s processing and approval.

5 For further enquiries on the upcoming controls, you may contact the following officers:

- Ms Felicia Lim at felicia_lim@nea.gov.sg
- Mr Thiraviyam Pandian at thiraviyam_pandian@nea.gov.sg
- Ms Tan Si Yan at tan_si_yan@nea.gov.sg

(This is a computer-generated circular and requires no signature.)

Annex A
Labelling Requirements for Paints for Outdoor Application and Industrial Use Containing ⩾ 0.01% Weight in Weight of Formaldehyde

The label must –

a) contain the following words clearly legible:
"**This paint contains formaldehyde.**
For application on outdoor areas or for industrial use only.
Do not apply on indoor areas."

b) conform to the following dimensions:
   i. where the capacity of the container does not exceed 3 litres – at least 52 × 74 millimetres;
   ii. where the capacity of the container exceeds 3 litres but does not exceed 50 litres – at least 74 × 105 millimetres;
   iii. where the capacity of the container exceeds 50 litres but does not exceed 500 litres – at least 105 × 148 millimetres;
   iv. where the capacity of the container exceeds 500 litres – at least 148 × 210 millimetres.

Annex B
Accepted Test Methods

a) The total in-can formaldehyde content in each paint product shall be determined for the colour tint containing the highest theoretical amount of formaldehyde that is predicted by the manufacturer.

b) The total in-can formaldehyde content in each paint product shall be measured in accordance with the methods indicated in the table below.

Type of paint	Total formaldehyde content measurement method	
	Extraction of formaldehyde	Analysis of extracted formaldehyde
Water-based	Acid-steam distillation [1]	(a) Photometric determination using chromotropic acid reagent [1], or (b) High performance liquid chromatographic (HPLC) determination using 2,4-dinitrophenylhydrazine reagent [3, 4].
Solvent-based	An appropriate method [2]	

[1] ISO 7012-2: Paints and varnishes — Determination of preservatives in water-dilutable coating materials, Part 2: Determination of in-can total formaldehyde; or equivalent.
[2] In the absence of international standard methods, NEA is amenable to appropriate test methods validated for extraction of total formaldehyde from solvent-based paints.
[3] GB/T 34683: Determination of formaldehyde content in water-borne coatings – High performance liquid chromatographic method; or equivalent.
[4] HPLC method shall be used if turbidity or coloration is observed in the distillate to avoid potential interference with the photometric method.

c) Test reports are considered valid if there are no changes in the paint formulations. NEA may also accept test reports for formaldehyde emissions on a case-by-case basis. Test methods for formaldehyde emissions shall be in accordance with SS 150:2021+A1:2024, ISO 16000-9:2024, ASTM D5116-17, EN 16516:2017+A1:2020 standard, or equivalent.

Information to be included in test reports

a) The product tested, packaging size, colour, manufacturer, manufacturing date, lot or batch number, and the date of opening for testing, where applicable, to ensure traceability.

b) In the case of water-based paint products, indication that the total formaldehyde is measured in accordance with ISO 7012-2:2025 or equivalent, along with a statement

confirming that the measurement method has been verified. The test conditions should also be stated and aligned with those specified in the test standards that are referenced to, including scale of the test and the volume of distillate collected. Any deviation from test standards should be highlighted and justified with valid reasons.

c) In the case of solvent-based paint products, a statement confirming that the in-house method has been validated.

d) A statement to confirm that the total formaldehyde is extracted through steam distillation with the steam source specified, and indicate if the extraction was conducted through manual glassware setup or commercial distillation apparatus.

e) Any unusual features or anomalies observed during the test, where applicable.

f) Test results should be reported in the correct units (e.g., total formaldehyde content in % w/w). If not, the laboratory should perform the necessary unit conversion.

g) The date, signature(s) of authorised personnel, and company stamp with company letter head.

Note: Documents detailing full method verification/ validation data, including parameters such as formaldehyde recovery, precision, linearity range, and limits of detection and quantification (non-exhaustive), shall be made available for NEA's review upon request. Recovery of formaldehyde shall be evaluated by spiking samples with formaldehyde-releasing compounds, and the names of the formaldehyde-releasing compounds used shall be reported. Method verification/validation data will be treated as confidential upon submission to NEA and need not be included in official test reports and certificates.

Annex C
List of Harmonized System ("HS") and Product Codes

HS Code	HS Description	Product Code	Product Description
32081090	Other paints based on polyesters in non-aqueous medium & solutions	PCDFIP211	Paints for interior application based on polyesters in non-aqueous medium & solutions with formaldehyde concentration equal to or exceeding 0.01% weight in weight
		PCDFIP311	Paints based on polyesters in non-aqueous medium & solutions with formaldehyde concentration below 0.01% weight in weight
		PCDFIP312	Paints for outdoor application or industrial use based on polyesters in non-aqueous medium & solutions with formaldehyde concentration equal to or exceeding 0.01% weight in weight
32082090	Other paints & varnishes in non-aqueous medium & solutions based on acrylic or vinyl polymers	PCDFIP211	Paints & varnishes for interior application in non-aqueous medium & solutions based on acrylic or vinyl polymers with formaldehyde concentration equal to or exceeding 0.01% weight in weight
		PCDFIP311	Paints & varnishes in non-aqueous medium & solutions based on acrylic or vinyl polymers with formaldehyde concentration below 0.01% weight in weight
		PCDFIP312	Paints & varnishes for outdoor application or industrial use in non-aqueous medium & solutions based on acrylic or vinyl polymers with formaldehyde concentration equal to or exceeding 0.01% weight in weight
32089090	Other paints in non-aqueous medium & solutions based on other polymers	PCDFIP211	Paints for interior application in non-aqueous medium & solutions based on other polymers for interior application

HS Code	HS Description	Product Code	Product Description
			with formaldehyde concentration equal to or exceeding 0.01% weight in weight
		PCDFIP311	Paints in non-aqueous medium & solutions based on other polymers with formaldehyde concentration below 0.01% weight in weight
		PCDFIP312	Paints for outdoor application or industrial use in non-aqueous medium & solutions based on other polymers with formaldehyde concentration equal to or exceeding 0.01% weight in weight
32091090	Other paints & varnishes of acrylic or vinyl polymers in aqueous medium	PCDFIP211	Paints & varnishes for interior application of acrylic or vinyl polymers in aqueous medium use with formaldehyde concentration equal to or exceeding 0.01% weight in weight
		PCDFIP311	Paints & varnishes of acrylic or vinyl polymers in aqueous medium with formaldehyde concentration below 0.01% weight in weight
		PCDFIP312	Paints & varnishes for outdoor application or industrial use of acrylic or vinyl polymers in aqueous medium with formaldehyde concentration equal to or exceeding 0.01% weight in weight
32099000	Other paints & varnishes of synthetic or chemically modified natural polymers in aqueous medium	PCDFIP211	Paints & varnishes for interior application of synthetic or chemically modified natural polymers in aqueous medium with formaldehyde concentration equal to or exceeding 0.01% weight in weight
		PCDFIP311	Paints & varnishes of synthetic or chemically modified natural polymers in aqueous medium

HS Code	HS Description	Product Code	Product Description
			with formaldehyde concentration below 0.01% weight in weight
		PCDFIP312	Paints & varnishes for outdoor application or industrial use of synthetic or chemically modified natural polymers in aqueous medium with formaldehyde concentration equal to or exceeding 0.01% weight in weight
32100099	Other paints & varnishes	PCDFIP211	Other paints & varnishes for interior application with formaldehyde concentration equal to or exceeding 0.01% weight in weight
		PCDFIP311	Other paints & varnishes with formaldehyde concentration below 0.01% weight in weight
		PCDFIP312	Other paints & varnishes for outdoor application or industrial use with formaldehyde concentration equal to or exceeding 0.01% weight in weight